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ORIGINAL RESEARCH

Transcatheter Aortic Valve Replacement in Patients With Systolic Heart Failure and Moderate Aortic Stenosis

TAVR UNLOAD

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ABSTRACT

BACKGROUND Neurohormonal modulation and afterload reduction are key for treatment of heart failure with reduced ejection fraction (HFrEF). In HFrEF patients with concomitant moderate aortic stenosis (AS), treatment with transcatheter aortic valve replacement (TAVR) may be complementary to guideline-directed medical therapy (GDMT).

OBJECTIVES This study sought to determine whether TAVR for moderate AS provides clinical benefit in patients with HFrEF on top of GDMT.

METHODS We performed an investigator-initiated, international, randomized controlled trial in patients with HFrEF on GDMT with moderate AS who were suitable for transfemoral TAVR with a balloon-expandable valve. Patients were randomized 1:1 to TAVR or clinical aortic stenosis surveillance (CASS) with aortic valve replacement upon progression to severe AS. The primary endpoint was the hierarchical occurrence of: 1) all-cause death; 2) disabling stroke; 3) disease-related hospitalizations and heart failure equivalents; and 4) change from baseline in the Kansas City Cardiomyopathy Questionnaire Overall Summary Score analyzed using the win ratio.

RESULTS From January 2017 to December 2022, 178 patients were randomized to TAVR (n = 89) or AS surveillance (n = 89). The mean age was 77 years, 20.8% were female, and 55.6% were in NYHA functional class III or IV. The median follow-up duration was 23 months (Q1-Q3: 12-33 months). A total of 38 (43%) patients in the CASS group (of whom 35 had progressed to severe AS) underwent TAVR at a median of 12 months postrandomization. TAVR was associated with wins in 47.6% of pairs, compared with 36.6% in the CASS group, resulting in a win ratio of 1.31 (95% CI: 0.91-1.88; $P = 0.14$). At 1 year, TAVR resulted in a greater improvement in the Kansas City Cardiomyopathy Questionnaire Overall Summary Score compared with the CASS group (12.8 ± 21.9 points vs 3.2 ± 22.8 points; $P = 0.018$).

CONCLUSIONS TAVR was not superior to AS surveillance for the primary hierarchical composite endpoint in patients with moderate AS and HFrEF on GDMT. Preemptive TAVR for moderate AS was safe and may provide clinically meaningful quality-of-life benefits. (JACC. 2025;85:878–890) © 2025 by the American College of Cardiology Foundation.

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Medical neurohormonal modulation and afterload reduction are key elements in the treatment of heart failure with reduced ejection fraction (HFrEF).[1] Degenerative aortic stenosis (AS) increases with aging and contributes to the overall valvuloarterial impedance in patients with HFrEF; treatment of AS may therefore represent a complementary target for afterload reduction in this population.[2] Aortic valve replacement (by either surgical or transcatheter means) is a guideline-recommended treatment for symptomatic severe AS, but not for moderate AS, with 1 exception: a Class IIb assessment that surgical aortic valve replacement (SAVR) may be reasonable for patients who require other open-heart surgery.[3] However, data from large echocardiographic databases have shown an association between moderate AS and long-term mortality.[4,5] Furthermore, over time, progression of AS may lead to maladaptive cardiac remodeling with hypertrophy and fibrosis that may irreversibly affect left ventricular (LV) systolic and diastolic function.[6] Cohort studies have reported high annual rates of mortality, heart failure (HF) hospitalizations, and valve interventions in patients with HFrEF and moderate AS.[7] A retrospective propensity-matched analysis also showed that patients with HFrEF had worse outcome in the presence of concomitant moderate AS.[8] A recent meta-analysis reported that moderate AS is associated with higher risk of death compared with patients with no or mild AS and that this association is stronger among patients with HFrEF.[9]

In recent years, transcatheter aortic valve replacement (TAVR) has emerged as a less invasive alternative to SAVR for treatment of patients with severe, symptomatic AS.[10,11] We hypothesized that TAVR in patients with HFrEF on guideline-directed medical therapy (GDMT) and moderate AS may provide incremental afterload reduction and clinically meaningful benefits as compared with clinical aortic stenosis surveillance (CASS) with aortic valve replacement upon progression to severe AS.[12] The TAVR UNLOAD (Transcatheter Aortic Valve Replacement to UNload the Left ventricle in patients with ADvanced heart failure; NCT02661451) trial was therefore designed to evaluate the impact of TAVR on clinical outcomes and quality of life in patients with HFrEF on GDMT and moderate AS.

SEE PAGE 891

METHODS

STUDY DESIGN AND OVERSIGHT. The design of the TAVR UNLOAD trial has been published previously but had to be modified during study enrollment as described in the statistical analysis section.[13] The TAVR UNLOAD trial was an investigator-initiated, international, randomized controlled, open label, superiority trial comparing transfemoral TAVR with CASS and aortic valve replacement upon progression to severe AS in symptomatic patients with HFrEF and moderate AS. The trial protocol and subsequent amendments were approved by Institutional Review Boards and corresponding health authorities at each participating site, and all patients gave written informed consent prior to study enrollment. The Cardiovascular Research Foundation was the trial sponsor. The academic investigators designed and supervised the trial, which was executed with the assistance of several clinical research organizations: the Cardiovascular Research Foundation Clinical Trial Center (New York, New York, USA), Cardialysis (Rotterdam, the Netherlands), Boston Biomedical Associates (Marlborough, Massachusetts, USA), and the Quebec Heart and Lung Institute Echo Core Laboratory (Quebec, Canada). An independent data and safety monitoring board provided oversight by periodically reviewing all reported serious adverse events.

ABBREVIATIONS AND ACRONYMS

AS = aortic stenosis

AVA = aortic valve area

CASS = clinical aortic stenosis surveillance

DSE = dobutamine stress echocardiogram

GDMT = guideline-directed medical therapy

HF = heart failure

HFrEF = heart failure with reduced ejection fraction

KCCQ = Kansas City Cardiomyopathy Questionnaire

KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire Overall Summary Score

LV = left ventricle/ventricular

LVEF = left ventricular ejection fraction

SAVR = surgical aortic valve replacement

TAVR = transcatheter aortic valve replacement

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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STUDY POPULATION. Local heart teams screened patients for eligibility based on routinely available clinical examination, echocardiography, coronary angiography, and multislice computed tomography assessment of aortic valve dimensions and calcifications, aorta and peripheral arteries. Patients had to be symptomatic (NYHA functional class II to IV), with a left ventricular ejection fraction (LVEF) between 20% and 50%, while receiving appropriate and stable GDMT for at least 1 month. Moderate AS was defined by an aortic valve area (AVA) >1.0 cm^2 and ≤1.5 cm^2 on a resting transthoracic echocardiogram. If AVA was ≤1.0 cm^2 at rest and low-flow AS was suspected, patients underwent a low-dose dobutamine stress echocardiogram (DSE). Patients with an AVA <1.0 cm^2 but with an indexed AVA of >0.6 cm^2/m^2 on either rest or DSE and patients with an AVA >1.5 cm^2 but with an indexed AVA <0.9 cm^2/m^2 on either rest or DSE were considered to have moderate AS. An independent echocardiography core laboratory confirmed moderate AS and LVEF prior to formal assessment by the screening committee. Anatomic suitability for transfemoral TAVR with a balloon-expandable valve (SAPIEN 3, Edwards Lifesciences) was assessed by multislice computed tomography. More details on the inclusion and exclusion criteria as well as the screening process are provided in Supplemental Appendix Section C.

RANDOMIZATION AND TREATMENT. Approved patients were randomly assigned in a 1:1 fashion to TAVR with a balloon expandable valve or CASS via a dedicated page in the electronic case report form. Patients randomized to TAVR were treated preferably within 2 weeks after randomization. Details of the TAVR procedure are presented in Supplemental Appendix Section D. Patients in the CASS group could only receive TAVR if AS had advanced from moderate to severe, as determined by echocardiography.

DATA COLLECTION AND FOLLOW-UP. Follow-up visits were planned at 1 month, 6 months, 1 year, 2 years, and once a year thereafter until December 31, 2023, with a maximum follow-up of 5 years. Clinical status and concomitant therapies were assessed at each visit, and laboratory tests and echocardiograms were performed according to the study schedule. In addition, we assessed health status at each visit using the Kansas City Cardiomyopathy Questionnaire (KCCQ), the Medical Outcomes Study 36-Item Short Form Health Survey, and the EuroQol 5 Dimension survey. The KCCQ is a HF-specific health status measure that has been validated both for patients with HFrEF and for patients with AS.[14,15] Scores on the Kansas City Cardiomyopathy Questionnaire Overall Summary Score (KCCQ-OSS) range from 0 to 100, with higher scores indicating better quality of life. Changes of >5, >10, and >20 at the individual patient level are considered clinically meaningful, moderate to large, and large to very large, respectively.[16] Functional status was assessed at each visit using the 6-minute walk test, while biological effects were measured by changes in N-terminal pro-B-type natriuretic peptide (or B-type natriuretic peptide) concentrations.

STUDY ENDPOINTS. The primary endpoint was the hierarchical occurrence of: 1) all-cause death; 2) disabling stroke; 3) disease-related hospitalizations and HF hospitalization equivalents; and 4) the change from baseline in the KCCQ-OSS. Disease-related hospitalizations included hospitalization for HF, symptomatic aortic valve disease, or nondisabling stroke. HF hospitalization equivalents included urgent visits to the emergency room or outpatient clinics, and intensification of HF medication. Details are provided in Supplemental Appendix Section E. The ranked key secondary endpoints were: 1) time to first occurrence of all-cause death, disabling stroke, or HF-related or aortic valve disease-related hospitalization or HF hospitalization equivalent; 2) the hierarchical occurrence of all-cause death, disabling stroke, or disease-related hospitalizations and equivalents; and 3) time to death from any cause. Endpoints were adjudicated by an independent clinical events committee. Additional endpoints and definitions are provided in Supplemental Appendix Section F.

STATISTICAL ANALYSIS. For assessment of the primary endpoint, the TAVR and CASS groups were compared across all possible patient pairs. Each pair of TAVR and clinical surveillance patients was assigned a win for TAVR, a win for clinical surveillance, or a tie based on the hierarchical occurrence of the primary endpoint, which was assessed at the longest common follow-up duration for the patient pair. For each pair, outcome assessment was restricted to events occurring within, and KCCQs recording within, the pair's common efficacy assessment time interval (ie, the shorter of the 2 available follow-up times of the respective 2 patients of the pair). We calculated the win ratio as the number of patient pairs that were a win for TAVR divided by the number of pairs that were a win for clinical surveillance. A 95% CI for the win ratio was calculated with the bootstrap method, and a 2-sided P value for superiority was obtained using the Finkelstein-Schoenfeld method (details of the pairwise analysis can be found in Supplemental Appendix Section G).

For the first and third key secondary endpoints, and all other time-to-first-event analyses, cause-specific HRs for TAVR vs CASS and associated 95% CIs were determined with the use of Cox proportional hazards models, and Kaplan-Meier curves were used to depict the occurrence of events over time. The analysis of the second key secondary endpoint used the same methodology as for the primary endpoint.

The original study design called for a sample size of 600 patients with the primary analysis on 1-year outcome data. In 2019, because of slow enrollment, the steering committee decided to prolong the 1-year follow-up period to include follow-up for all patients until an event-driven calendar-based end-of-study date. Under updated assumptions (144 primary endpoint events among 302 randomized patients), we estimated that the study would retain >80% power for the Finkelstein-Schoenfeld analysis of the primary endpoint and ~80% power for the first secondary endpoint (major adverse cardiovascular and cerebrovascular events).[17] In mid-2022, due to funding constraints and continued slower than anticipated enrollment, the steering committee decided to end screening for enrollment on December 31, 2022, and to conduct the primary analysis based on patient outcomes that occurred before December 31, 2023, or 1 year after randomization, whichever came last.

The statistical analyses of the primary and ranked key secondary endpoints were conducted with the use of adjudicated outcome events and according to the intention-to-treat principle in all patients who had undergone randomization, regardless of whether they adhered to their assigned treatment strategy. All analyses included clinical endpoints and, when applicable, KCCQ outcomes that occurred between randomization and the current analysis's date of censoring, which was December 31, 2023, 1 year after randomization, 5 years after randomization, or the latest date of known endpoint status, whichever was applicable. The statistical analyses of the original primary and ranked key secondary endpoints at 1-year are presented as exploratory analyses.

RESULTS

PATIENTS. Between January 2017 and December 2022, 279 patients at 66 institutions in the United States, the Netherlands, and Austria provided informed consent and underwent screening investigations. A total of 178 patients were accepted for randomization by the central screening committee (**Figure 1**), and 89 patients were randomly assigned to TAVR or CASS, respectively. Details of patient flow including withdrawal of consent, loss to follow-up, and the performance of aortic valve replacement in the clinical surveillance group are summarized in **Figure 1**. A total of 3 patients in the TAVR group and 5 in the CASS group either withdrew or were lost to follow-up by 1 year; these counts increased to 4 and 11, respectively, at maximum follow-up. The median duration of follow-up was 23 months (Q1-Q3: 12-33 months). The outcome status at scheduled end-of-study was available in 163 (91.6%) patients with premature censoring of follow-up at the time of last known clinical status in the remaining 15 patients.

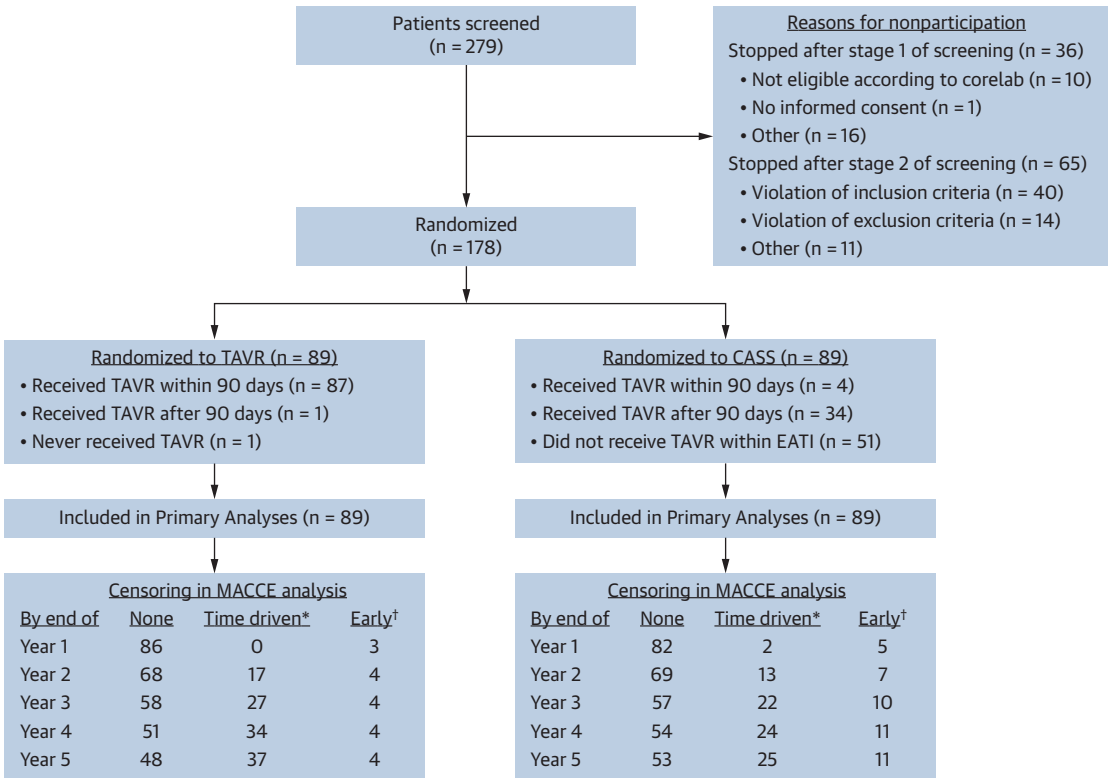
The baseline characteristics of the patients in the 2 trial groups were well matched (**Table 1**). The mean age was 77.4 ± 7.2 years, 20.8% were women, the Society of Thoracic Surgeons Predicted Risk of Mortality was 4.4% ± 3.4%, and 27% were frail. The HF etiology was ischemic in 57 (64%) of 89 patients in the TAVR group and 61 (68.5%) of 89 in the CASS group. Overall, 55.6% were in NYHA functional class III or IV, the mean KCCQ-OSS was 55.8 ± 23.1 points, and the mean 6-minute walk distance was 232 ± 163 m. The mean LVEF was 39% ± 9.6%, the mean AVA was 1.2 ± 0.2 cm^2, and the mean aortic valve gradient was 21.9 ± 6.8 mm Hg. DSE was performed in 80 (44.9%) of 178 patients in the study, 38 in the TAVR group and 42 in the CASS group. There were no cases of severe mitral, aortic, or tricuspid regurgitation at baseline. At baseline, most participants were on beta-blockers, diuretics, and renin-angiotensin system blockers; 47.4% were receiving mineralocorticoid receptor antagonists and 13.3% were receiving cardiac glycosides (**Table 1**).

PROCEDURES AND FOLLOW-UP. TAVR was performed in 88 (99%) of the patients in the TAVR group, with successful implantation in 87 patients (1 converted to SAVR), at a median of 13 days (Q1-Q3: 7-15 days) after randomization. During the first 30 days postprocedure, there were no deaths in the TAVR group. One (1.1%) patient experienced a disabling stroke, 1 (1.1%) underwent pericardial drainage for pericardial effusion (Bleeding Academic Research Consortium 3), 1 (1.1%) was hospitalized for acute HF, and 1 (1.1%) developed acute kidney injury. There were no major vascular complications, coronary obstructions, or episodes of new onset atrial fibrillation. Four (4.5%) patients required permanent pacemaker implantation for high-degree atrioventricular block (Supplemental Table 1).

In the CASS group, 38 (43%) patients underwent TAVR at a median of 12 months after randomization (Q1-Q3: 6-20 months), of whom 35 (92%) had

JACC VOL. 85, NO. 9, 2025
MARCH 11, 2025:878-890

FIGURE 1 CONSORT Diagram



Study flow diagram showing the flow of participants through each stage of the TAVR UNLOAD (Transcatheter Aortic Valve Replacement to UNload the Left ventricle in patients with ADvanced heart failure) trial. *Censoring because of late randomization and/or early visit. †Because of withdrawal of informed consent or loss to follow-up. CASS = clinical aortic stenosis surveillance; EATI = efficacy assessment time interval; MACCE = major adverse cardiovascular and cerebrovascular events; TAVR = transcatheter aortic valve replacement.

progression to severe AS. A total of 17 (49%) of 35 patients experienced a disease-related hospitalization or equivalent before diagnosis of severe AS and indication for TAVR, while the remaining 18 (51%) underwent TAVR exclusively due to progression to severe AS on a follow-up echocardiogram. Three (8%) patients underwent TAVR with only moderate AS. Of these 3 patients, 2 were hospitalized for decompensated HF before TAVR (at 7 months and at 11 months after randomization) and 1 was an early crossover (at 37 days) after being hospitalized due to chest pain that the independent clinical events committee attributed to coexisting coronary artery disease. One additional patient (who refused TAVR) was treated with balloon aortic valvuloplasty after multiple HF events.

CLINICAL OUTCOMES. In the longest follow-up analysis of the hierarchical composite primary endpoint (based on major adverse cardiovascular and cerebrovascular events and KCCQ-OSS), comparison of patient pairs demonstrated wins for TAVR in 47.6% of pairs, CASS in 36.3%, and ties in 16.1%, resulting in a win ratio of 1.31 (95% CI: 0.91-1.88; $P = 0.143$). TAVR was associated with more wins for each component of the hierarchical composite endpoint, but these differences were not statistically significant (**Figure 2**). In the 1-year ranked ordinal comparison, TAVR led to more wins than CASS (48% vs 30.9%; win ratio: 1.55; 95% CI: 1.04-2.31). There were no significant differences between TAVR and CASS in the individual components of the ordinal composite endpoints (including all-cause mortality) either at longest follow-up or at 1 year (**Table 2**, **Figure 3**). Mortality was most often attributed to cardiovascular causes (Supplemental Listing 1). In the first year, 24 of 89 patients in the TAVR group and 28 of 89 patients in the CASS group experienced at least 1 disease-related hospitalization or HF hospitalization

TABLE 1 Characteristics of the Patients at Baseline

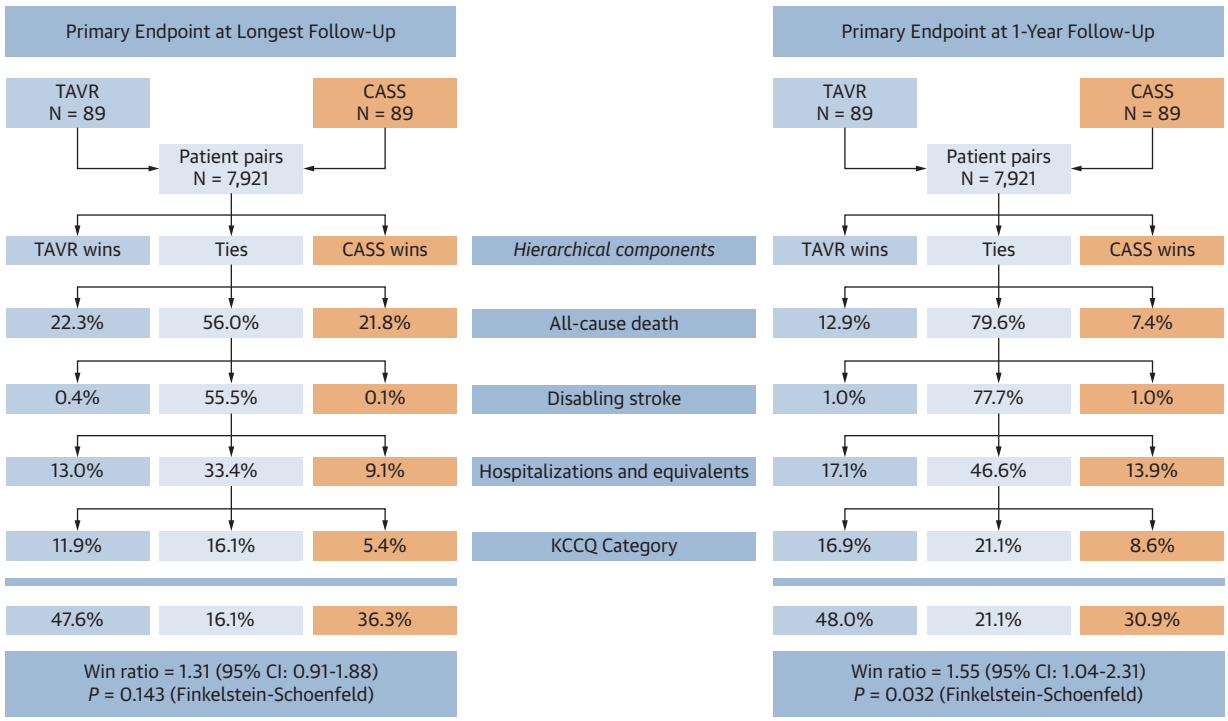
	TAVR (n = 89)	CASS (n = 89)	All (N = 178)
Age, y	76.6 ± 7.7	78.2 ± 6.6	77.4 ± 7.2
Female	16 (18.0)	21 (23.6)	37 (20.8)
Body mass index, kg/m^2	28.9 ± 5.5	28.9 ± 5.7	28.9 ± 5.6
STS PROM score, %[a]	4.0 ± 2.6	4.8 ± 4.0	4.4 ± 3.4
NYHA functional class			
I	1 (1.1)	1 (1.1)	2 (1.1)
II	34 (38.2)	43 (48.3)	77 (43.3)
III	51 (57.3)	42 (47.2)	93 (52.2)
IV	3 (3.4)	3 (3.4)	6 (3.4)
Any HFH during the past 12 mo	37 (42.0)	42 (47.7)	79 (44.9)
Coronary artery disease	66 (74.2)	70 (78.7)	136 (76.4)
Previous myocardial infarction	40 (44.9)	40 (44.9)	80 (44.9)
Previous coronary revascularization	61 (68.5)	65 (73.0)	126 (70.8)
Previous stroke	8 (9.0)	8 (9.0)	16 (9.0)
Pre-existing implantable devices			
Pacemaker	11 (12.4)	6 (6.7)	17 (9.6)
ICD	31 (34.8)	31 (34.8)	62 (34.8)
CRT	10 (11.2)	0 (0.0)	10 (5.6)
Peripheral vascular disease	19 (21.3)	20 (22.5)	39 (21.9)
Chronic obstructive pulmonary disease	14 (15.7)	20 (22.5)	34 (19.1)
Creatinine >2 mg/dL	7 (8.2)	4 (4.7)	11 (6.4)
Atrial fibrillation	40 (45.5)	50 (56.2)	90 (50.8)
Frailty condition[b]	19 (27.5)	16 (26.2)	35 (26.9)
Aortic valve area (baseline), cm^2	1.1 ± 0.2	1.2 ± 0.2	1.2 ± 0.2
Aortic valve area (baseline or dobutamine stress), cm^2	1.2 ± 0.2	1.3 ± 0.2	1.2 ± 0.2
Mean aortic valve gradient (baseline), mm Hg	19.1 ± 6.0	18.5 ± 5.9	18.8 ± 5.9
Mean aortic valve gradient (baseline or dobutamine stress), mm Hg	21.9 ± 6.9	21.9 ± 6.7	21.9 ± 6.8
Left ventricular ejection fraction, %	39.3 ± 9.6	38.8 ± 8.6	39.0 ± 9.1
Left ventricular end-diastolic volume, mL	180 ± 50.3	183 ± 56.5	181.6 ± 53.4
Left ventricular end-diastolic diameter, cm	6.9 ± 7.4	6.3 ± 5.3	6.6 ± 6.4
Moderate or moderate-severe regurgitation[c]			
Aortic	3 (3.4)	2 (2.4)	5 (2.9)
Mitral	6 (6.9)	6 (7.2)	12 (7.1)
Tricuspid	12 (15.0)	12 (15.0)	24 (15.0)
KCCQ-OSS at baseline	57.1 ± 24.3	54.5 ± 21.8	55.8 ± 23.1
6-min walk distance at baseline, m	236 ± 154.9	229 ± 172.4	232.5 ± 163.2
Heart failure medications at baseline			
ACE inhibitors/angiotensin 2 receptor blockers/sacubitril-valsartan	63/87 (72.4)	63/88 (71.6)	126/175 (72.0)
Beta-blockers	77/87 (88.5)	75/89 (84.3)	152/176 (86.4)
Mineralocorticoid receptor antagonists	46/87 (52.9)	36/86 (41.9)	82/173 (47.4)
Diuretics	70/88 (79.5)	77/89 (86.5)	147/177 (83.1)
Cardiac glycosides	12/86 (14.0)	11/87 (12.6)	23/173 (13.3)

Values are mean ± SD, n (%), or n/N (%). Data on aortic valve area were available for 88 patients in the TAVR group and 87 patients in the CASS group: mean aortic valve gradient was 89 mm Hg and 88 mm Hg, respectively; frailty condition was 69 and 61, respectively; left ventricular ejection fraction was 88% and 88%, respectively; KCCQ-OSS was 85 and 81, respectively; and 6-minute walk distance was 85 m and 81 m, respectively. Dobutabine stress echocardiogram data were available in 76 (43%) of 175 aortic valve area values and 80 (45%) of 177 aortic valve mean gradient values. [a]STS PROM scores range from 0% to 100%, with higher scores indicating a greater risk of death within 30 days after the procedure. [b]Frailty was defined as the presence of 3 or more of the following criteria: grip strength of <18 kg, 5-m walk-test time of more than 6 seconds, serum albumin level of <3.5 g/dL, and Katz Index of Independence in Activities of Daily Living total score of 4 or less (with scores ranging from 0 to 6, with higher scores indicating greater independence in performing activities of daily living). [c]There were no cases of severe mitral regurgitation, aortic regurgitation, or tricuspid regurgitation at baseline.

ACE = angiotensin-converting enzyme; CASS = clinical aortic stenosis surveillance; CRT = cardiac resynchronization therapy; HFH = heart failure hospitalization; ICD = implantable cardioverter-defibrillator; KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire Overall Summary Score; PROM = Predicted Risk of Mortality; STS = Society of Thoracic Surgeons; TAVR = transcatheter aortic valve replacement.

JACC VOL. 85, NO. 9, 2025
MARCH 11, 2025:878–890

FIGURE 2 Primary Endpoint at Longest and 1-Year Follow-Up



Hierarchical composite primary endpoint demonstrating wins for TAVR in 47.6% of pairs, CASS in 36.3%, and ties in 16.1%, resulting in a win ratio of 1.31 (95% CI: 0.91-1.88; $P = 0.143$) at longest follow-up. In the 1-year ranked ordinal comparison, TAVR led to more wins than CASS (48% vs 30.9%; win ratio: 1.55; 95% CI: 1.04-2.31). KCCQ = Kansas City Cardiomyopathy Questionnaire; other abbreviations as in **Figure 1**.

equivalent. Details of these events are provided in Supplemental Table 2.

ECHOCARDIOGRAPHY. In the TAVR group, the AVA increased from 1.1 ± 0.2 cm^2 to 1.8 ± 0.6 cm^2 ($P < 0.0001$), the mean gradient fell from 19.1 ± 6.0 mm Hg to 9.3 ± 5.8 mm Hg ($P < 0.0001$), and the LVEF increased from 39.3% ± 9.6% to 44.2% ± 9.7% ($P = 0.002$) between baseline and 1-year follow-up. In the CASS group, among patients who did not undergo TAVR by 1-year follow-up, the AVA remained within the range of moderate AS (1.2 ± 0.2 cm^2 vs 1.2 ± 0.3 cm^2; $P = 0.79$), the mean gradient increased from 18.2 ± 5.8 mm Hg to 21.1 ± 8.6 mm Hg ($P = 0.038$), and the LVEF increased from 38.3% ± 8.1% to 43.5% ± 10.0% ($P = 0.0008$). In the CASS group, moderate AS progressed to severe AS in 16 patients at 1 year, to 13 additional patients at 2 years, to 5 more patients at 3 years, and to 1 patient at 40 months of follow-up (Supplemental Figure 1).

QUALITY OF LIFE. Among the TAVR group, the KCCQ-OSS increased immediately after TAVR, which persisted through 1-year follow-up (**Figure 4**). When compared with the CASS strategy, TAVR demonstrated an early health status benefit (as assessed by the KCCQ-OSS), which was apparent at 30 days and persisted through 1-year follow-up (12.8 ± 21.9 points vs 3.2 ± 22.8 points; $P = 0.018$). Although follow-up KCCQ-OSS remained higher with TAVR at 2- and 3-year follow-up, these differences were no longer statistically significant. When post-TAVR KCCQ measurements from control arm patients who converted to guideline-indicated TAVR were excluded from the analysis for exploratory purposes, KCCQ-OSS did not improve in the CASS group, leading to significant differences between groups at both 1-year and the longest available follow-up (**Figure 4**).

At 1 year, the KCCQ score was significantly higher in the TAVR group compared with the CASS group in the intention-to-treat population; 67.2% of patients in the TAVR group vs 42.4% of patients in the CASS cohort had an improvement of KCCQ ≥5 ($P = 0.007$ for chi-square test). A total of 60% of patients in the TAVR group were alive and experienced improvement of KCCQ ≥5 points vs 35.2% patients in the CASS group ($P = 0.003$) (**Table 3**). At longest follow-up,

TABLE 2 Primary and Secondary Clinical Endpoints

Pairwise Comparison Endpoint	TAVR Win, % (n = 89)	CASS Win, % (n = 89)	Win Ratio (95% CI)	*P* Value[a]
At longest follow-up				
Hierarchical composite endpoint (events and KCCQ)	47.6	36.3	1.31 (0.91-1.88)	0.14
Hierarchical composite endpoint (events)	35.7	30.9	1.15 (0.76-1.76)	0.51
At 1-y follow-up				
Hierarchical composite endpoint (events and KCCQ)	48.0	30.9	1.55 (1.04-2.31)	0.032
Hierarchical composite endpoint (events)	31.1	22.3	1.39 (0.82-2.35)	0.22

Time-to-Event Endpoint	n (%)	Event Rate[b]	n (%)	Event Rate[b]	HR (95% CI)	*P* Value[c]
At longest follow-up						
Major adverse cardiovascular and cerebrovascular events	46 (51.7)	33.0	51 (57.3)	41.6	0.83 (0.56-1.24)	0.37
All-cause death	35 (39.3)	18.6	35 (39.3)	19.4	0.98 (0.61-1.56)	0.92
Any stroke	2 (2.2)	1.1	6 (6.7)	3.4	0.32 (0.06-1.58)	0.16
Disease-related hospitalizations or HFH equivalents	34 (38.2)	24.4	40 (44.9)	31.7	0.80 (0.51-1.27)	0.35
At 1-y follow-up						
Major adverse cardiovascular and cerebrovascular events	26 (29.2)	29.7	33 (37.1)	38.8	0.75 (0.45-1.26)	0.27
All-cause death	8 (9.0)	9.3	12 (13.5)	14.3	0.60 (0.25-1.48)	0.26
Any stroke	1 (1.1)	1.1	4 (4.5)	5.1	0.23 (0.03-2.09)	0.16
Disease-related hospitalizations or HFH equivalents	24 (27.0)	27.5	28 (31.5)	33.9	0.83 (0.48-1.43)	0.49

[a]Finkelstein and Schoenfeld *P* value. [b]Event rates at longest follow-up are incidence rates per 100 patient-years of follow-up; event rates at 1-year follow-up are Kaplan-Meier estimates (%). [c]Log-rank *P* value.

KCCQ = Kansas City Cardiomyopathy Questionnaire; other abbreviations as in Table 1.

61.2% of patients in the TAVR group vs 43.2% of patients in the CASS cohort had an improvement of KCCQ ≥5 points ($P = 0.024$ for chi-square test).

OTHER OUTCOMES. At 1-year follow-up, there were no significant differences in changes relative to baseline for generic health status as assessed by the 36-Item Short Form Health Survey physical summary score (TAVR vs CSS: 1.4 ± 5.5 vs 0.2 ± 5.6; $P = 0.23$) or mental summary score (TAVR vs CSS: −0.8 ± 7.3 vs 0.3 ± 5.7; $P = 0.38$). Similar results were seen for the EuroQol index ($P = 0.34$), 6-minute walk distance (TAVR vs CSS: mean change from baseline −5.1 ± 157.4 m vs −30 ± 222.7 m; $P = 0.52$) and N-terminal pro-B-type natriuretic peptide (TAVR vs CSS: 105 ± 1,853 pg/mL vs −1,229 ± 7,874 pg/mL; $P = 0.31$) or B-type natriuretic peptide (TAVR vs CSS: −92 ± 306 pg/mL vs 57 ± 122 pg/mL; $P = 0.30$) (Supplemental Table 3).

DISCUSSION

The TAVR UNLOAD trial is the first randomized controlled trial evaluating TAVR for moderate AS in patients with HFrEF on GDMT. The challenges of slow enrollment and the need for protocol modifications compel caution for the overall study interpretation. The key findings can be summarized as follows: 1) there was no difference in the primary hierarchical composite endpoint at a median follow-up of 23 months; 2) preemptive TAVR for moderate AS in patients with HFrEF on GDMT was safe; 3) TAVR resulted in more wins in the primary hierarchical composite endpoint at 1 year follow-up driven by improvement in quality of life; 4) TAVR resulted in clinically meaningful improvements in quality of life at 1 year compared with CASS; and 5) during the trial, 43% of the CASS group underwent TAVR at a median of 12 months postrandomization, predominantly because of disease progression to severe AS.

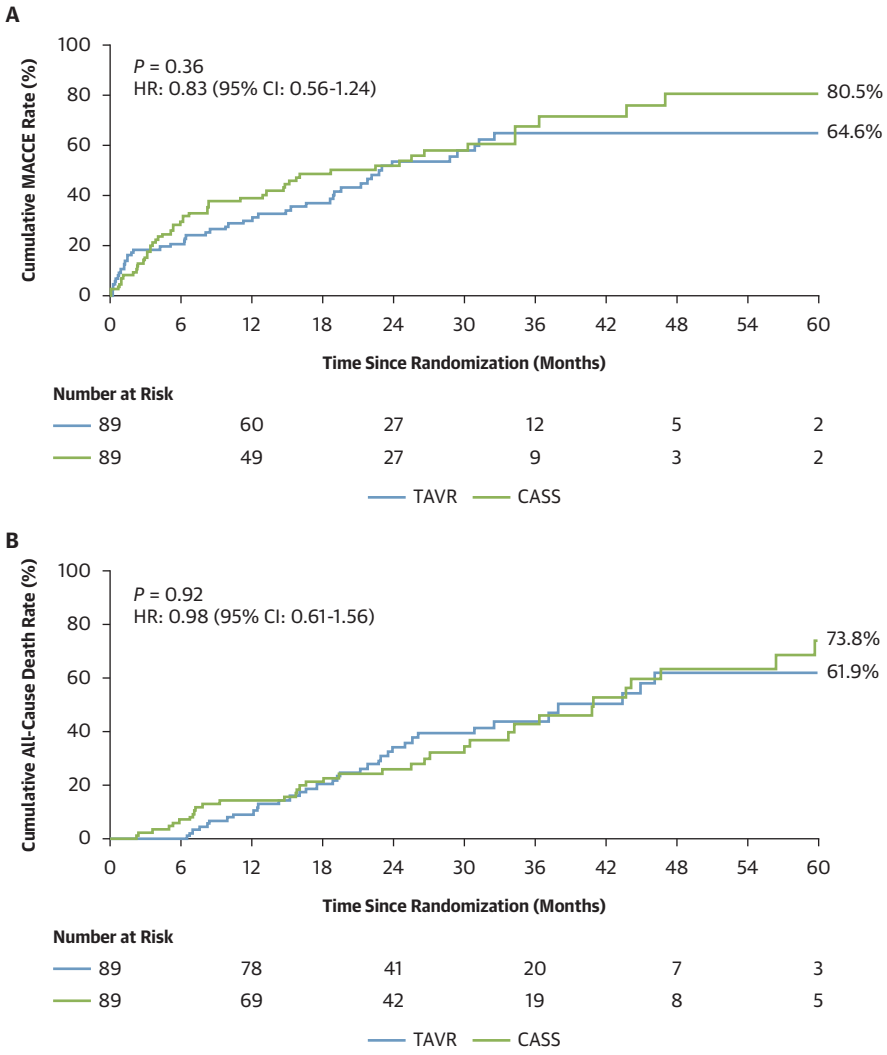
Patients in the TAVR UNLOAD trial had an advanced HFrEF phenotype. Overall, 56% of patients were in NYHA functional class III or IV, 44% had experienced 1 or more HF hospitalizations in the year prior to study enrollment, and the mean KCCQ-OSS was 57 points at baseline (roughly equivalent to NYHA functional class III).

We identified no safety concerns with TAVR in our study cohort—a population that was largely at intermediate operative risk (mean Society of Thoracic Surgeons Predicted Risk of Mortality of 4.4%). Procedure-related complications were infrequent and reflect the established safety profile of TAVR in contemporary trials.[18,19]

There was no difference in the primary hierarchical endpoint at latest follow-up between TAVR and CASS. The clinical event rates were higher than previously reported in retrospective studies on prognostic

JACC VOL. 85, NO. 9, 2025
MARCH 11, 2025:878–890

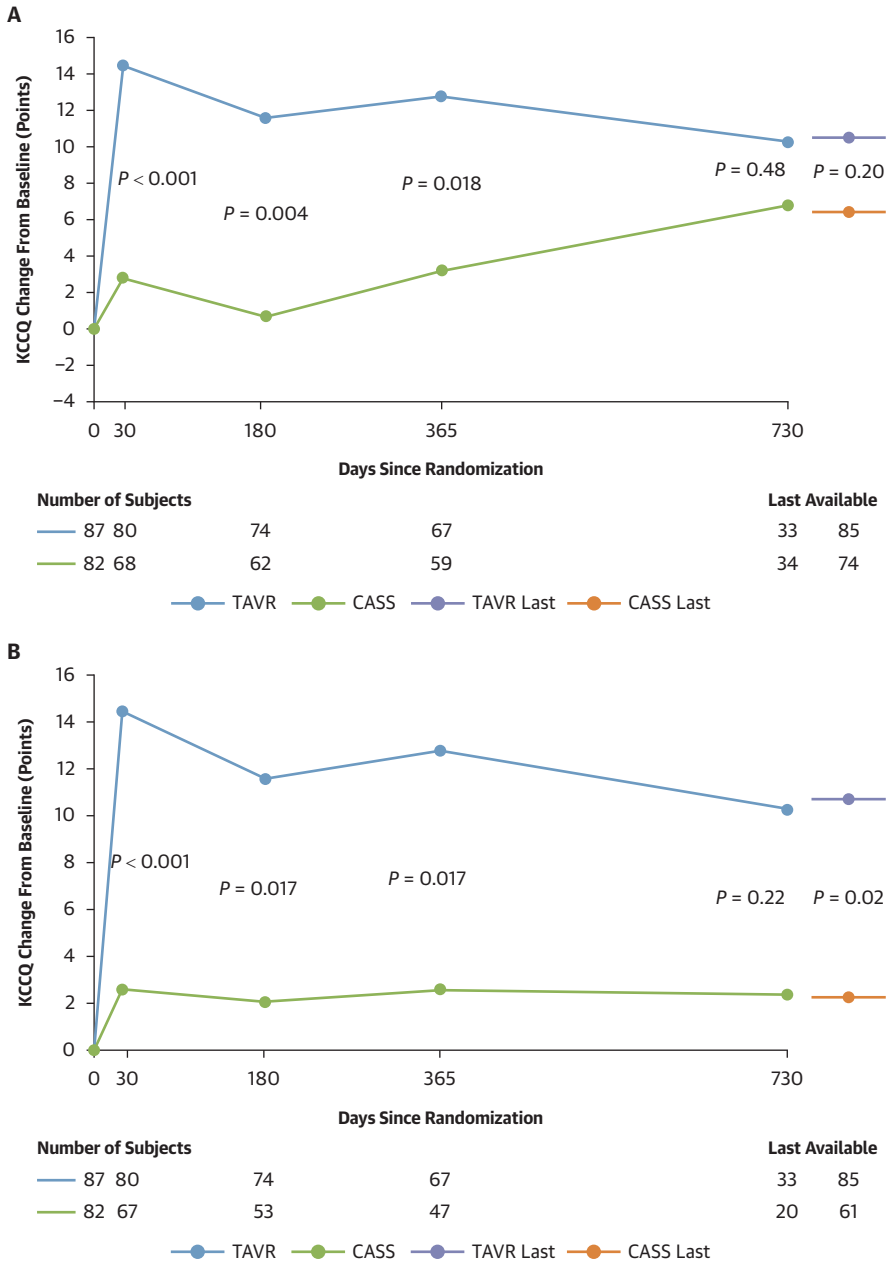
FIGURE 3 Kaplan-Meier Curves for the Key Secondary Endpoints



Kaplan-Meier curve for (A) major adverse cardiovascular and cerebrovascular events (MACCE) and (B) all-cause death. MACCE is a composite of all-cause death, stroke, disease-related hospitalizations, or heart failure hospitalization equivalents. Abbreviations as in Figure 1.

implications of moderate AS in patients with HFrEF, most likely because the patients in the TAVR UNLOAD trial were older and more symptomatic.[7] Importantly, the results of the TAVR UNLOAD trial may question several retrospective studies suggesting that correction of moderate AS in patients with HFrEF would lead to improved survival.[8,12] The prevalence of multiple comorbidities (including high rates of coronary artery disease, kidney disease, atrial fibrillation, frailty, and moderate-severe tricuspid regurgitation) and the predominantly ischemic nature of the cardiomyopathy in our study population may have been a contributing factor to explain the lack of survival benefit of TAVR. We also observed that the need for guideline-indicated TAVR in the control group occurred earlier than expected, as indicated by a median time to TAVR of 12 months; roughly 1 of 6 patients in the control arm required TAVR within 1 year. Based on previous studies, the expected rate of progression of calcific AS is 0.08 to 0.14 cm^2 per year.[20,21] The pattern of disease progression in the TAVR UNLOAD trial suggests that more frequent

FIGURE 4 KCCQ Changes Over Time



(A) All available Kansas City Cardiomyopathy Questionnaire (KCCQ) measurements over time and (B) excluding KCCQ measurements performed after TAVR in the CASS group. This analysis was exploratory and was not pre-specified. Comparisons of the last available KCCQ for each patient are shown at the right of each panel, and are nonsignificant with all data ($P = 0.20$), but significant without post-TAVR values in the CASS group ($P = 0.02$). Abbreviations as in Figure 1.

echocardiographic follow-up (eg, at 6-month intervals) may be warranted in patients with HFrEF and moderate AS to assure timely recognition of progression to severe AS.

We found a moderate to large improvement in health status with TAVR (+12.8 ± 21.9) that was significantly greater than that seen with CASS at 1 year. Given the open label trial design, we cannot rule

JACC VOL. 85, NO. 9, 2025
MARCH 11, 2025:878-890

TABLE 3 KCCQ Changes and Responder Analysis

	TAVR (n = 89)	CASS (n = 89)	Difference (95% CI)	*P* Value
Change in KCCQ at longest follow-up	85	74		
Mean change	10.9 ± 22.0	6.1 ± 25.0	4.8 (−2.6 to 12.2)	0.20
Change in 3 categories				0.037
≥5-point improvement	52 (61.2)	32 (43.2)		
<5-point change	15 (17.6)	20 (27.0)		
≥5-point worsening	18 (21.2)	22 (29.7)		
Change in KCCQ at 1 y	67	59		
Mean change	12.8 ± 21.9	3.2 ± 22.8	9.6 (1.7-17.4)	0.018
Change in 3 categories				0.002
≥5-point improvement	45 (67.2)	25 (42.4)		
<5-point change	12 (17.9)	12 (20.3)		
≥5-point worsening	10 (14.9)	22 (37.3)		
Responder at 1 y	75	71		
Alive and KCCQ improvement of ≥5 points	45 (60.0)	25 (35.2)	24.8% (9.1%-40.5%)	0.003
Alive and KCCQ ≥60 points and improvement of ≥10 points	48 (64.0)	25 (35.2)	28.8% (13.3%-42.5%)	<0.001

Values are n, mean ± SD, or n (%), unless otherwise indicated.
Abbreviations as in Tables 1 and 2.

out potential placebo effects. Nonetheless, the magnitude of the observed difference, even when accounting for conversions to TAVR in the clinical surveillance group, as well as the persistent duration of benefit (>1 year) merits further research. This difference most likely reflects unloading of the LV after TAVR, as reflected by substantial increases in AVA and reductions in transvalvular gradient. Of note, while the magnitude of improvement in the KCCQ at 1 year after TAVR in the TAVR UNLOAD trial was somewhat lower than the 16- to 25-point improvements seen in the pivotal randomized trials comparing TAVR with medical therapy or SAVR for severe AS,[20-23] they are quite similar to those observed in trials of transcatheter therapies for other HF populations such as patients with severe secondary mitral regurgitation and tricuspid regurgitation. In the COAPT (Cardiovascular Outcomes Assessment of the MitraClip Percutaneous Therapy for Heart Failure Patients with Functional Mitral Regurgitation) trial, for example, the 1-year benefit of mitral transcatheter edge-to-edge repair compared with GDMT alone in patients with HFrEF and secondary mitral regurgitation was 14.5 points.[24] We suspect that the smaller benefit of TAVR in the setting of LV systolic dysfunction reflects the fact that TAVR only corrects 1 cause of impaired status in these patients (aortic stenosis), whereas the extent of LV dysfunction largely persists.

The health status benefit of TAVR in the TAVR UNLOAD trial should be considered an exploratory finding that requires verification/replication in future studies, especially since we observed that 36-Item Short Form Health Survey and EuroQol were not impacted by the intervention. The lack of improvement in generic health status measures is not particularly surprising, since multiple studies validating the KCCQ have shown it to be significantly more sensitive than generic health status instruments for detecting clinically relevant changes in HF populations.[14,15] Importantly, patients with HFrEF and moderate AS showed clinical benefit after TAVR in addition to GDMT, as evidenced by the 60% of patients in the TAVR group who were alive with meaningful KCCQ improvements at 1 year vs the 35.2% of patients in the CASS group. This 60% responder rate in the TAVR UNLOAD trial mirrors the expected responder rate for TAVR in patients with severe AS at intermediate operative risk.[25] The high conversion rate to TAVR in the CASS group (43%) in our study likely explains the attenuation of the relative health status benefit for the TAVR group beyond 1 year.

STUDY LIMITATIONS. The TAVR UNLOAD trial underwent multiple study design changes and remained underpowered to discern significant changes in the primary endpoint (eg, <70% power for the Finkelstein-Schoenfeld test based on pretrial simulations).[17] Our findings are therefore hypothesis-generating. The trial experienced a slow enrollment rate because of the absence of an established referral pathway for patients with moderate AS from secondary care hospitals to the tertiary care institutions that participated in the TAVR UNLOAD trial and

during the COVID-19 pandemic, echocardiography studies were reduced in priority, which may have hampered the detection of moderate AS in patients with HFrEF. The steering committee had implemented changes in the protocol to reduce the target sample size including a change in the initially planned 1-year follow-up to the longest available follow-up. Although the study was underpowered to adequately test the clinical benefits of TAVR in patients with HFrEF on GDMT with moderate AS, the outstanding procedural safety of TAVR in these patients and the immediate effects on quality of life may favor preemptive TAVR for select symptomatic patients. The CASS group progressed relatively quickly to severe AS, resulting in a higher than expected number of patients receiving TAVR during the study period, which may have diminished intergroup comparisons. Utilizing the cardiac damage framework may identify a broader patient phenotype that experiences the afterload imposed by moderate AS and may benefit from upstream TAVR.[26,27] This concept is being studied in the randomized PROGRESS (Prospective, Randomized, Controlled Trial to Assess the Management of Moderate Aortic Stenosis by Clinical Surveillance or Transcatheter Aortic Valve Replacement; NCT04889872) trial and EXPAND TAVR II Pivotal Trial (NCT05149755).

During the TAVR UNLOAD trial, GDMT changed with the introduction of neprilysin/valsartan and sodium-glucose cotransporter 2 inhibitors. Adoption of these newer HF drugs was limited in the TAVR UNLOAD trial. Whether greater use of these newer drugs would have impacted our findings is unknown, but we believe that any impact on the between group comparisons would be modest owing to the distinct mechanism of benefit of TAVR. Finally, the relatively small sample size precluded heterogeneity testing.

CONCLUSIONS

TAVR did not affect the primary hierarchical composite endpoint in patients with HFrEF on GDMT as compared with CASS. Preemptive TAVR for moderate AS was safe and may provide early and clinically meaningful quality-of-life benefits.

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APPENDIX For an expanded Methods section, supplemental tables, and a supplemental figure, please see the online version of this paper.